

Research Interests

Statistical physics of disordered systems, non-equilibrium dynamics, creep motion, depinning, thermally activated failure, and amorphous plasticity.

Education

- 2024–Present **Ph.D. in Theoretical Statistical Physics**, *Laboratoire de Physique Théorique et Modèles Statistiques (LPTMS) and Laboratoire de Physique et Mécanique des Milieux Hétérogènes (PMMH), CNRS and Université Paris-Saclay*, Paris, France
Thesis title: “Creep dynamics in driven disordered systems”
Supervisors: Alberto Rosso (LPTMS) and Damien Vandembroucq (PMMH)
- 2021–2023 **Master of Science (M.Sc.) in Physical Sciences – Theoretical Physics**, *Università degli Studi di Pavia*, Pavia, Italy
Thesis title: “Two interacting spins driven out of equilibrium and in presence of dissipation”
External Supervisor: Alberto Rosso, Directeur de Recherche CNRS
Internal Supervisor: Prof. Paolo Perinotti
- 2023 **Erasmus Traineeship**, *Laboratoire de Physique Théorique et Modèles Statistiques (LPTMS), Université Paris-Saclay*, Paris, France
Research internship devoted to Master’s thesis work
Supervisor: Alberto Rosso, Directeur de Recherche CNRS
- 2017–2021 **Bachelor of Science (B.Sc.) in Physics**, *Università degli Studi di Pavia*, Pavia, Italy
Essay title: “Introduzione alla Fissione Nucleare” – “Nuclear Fission: an Introduction”
Supervisor: Chiar.ma Prof.ssa Barbara Pasquini
- 2017 **High School Diploma – Classical Studies**, *Liceo Statale “Giuseppe Novello”*, Codogno, Italy
Curriculum focused on Latin, Ancient Greek, Italian Literature, History, Philosophy, and Art History

Publications

- Preprint G. Russo, E. E. Ferrero, A. B. Kolton, A. Rosso, and D. Vandembroucq, *Two length scales in finite-temperature creep of an elastic interface*, arXiv:2604.17600, 2026.

Visiting Positions

- December 2024 **Visiting Scholar**, *Instituto Balseiro – Centro Atómico Bariloche*, Bariloche, Argentina
Research visit during Ph.D. studies in collaboration with Alejandro B. Kolton (Instituto Balseiro) and E. Ezequiel Ferrero (CONICET)

Conferences and seminars

- July 2026 **IRP ReMCoM2026 Kick-off Meeting**, Grenoble, Auvergne-Rhône-Alpes, France
Invited talk: “Creep motion of driven disordered systems”
- July 2026 **6th PSL Soft and Living Matter Days**, Paris, Île-de-France, France
Contributed talk: “Creep motion of an elastic line: dynamical heterogeneities”
- June 2026 **EDPIF Journée Scientifique**, Paris, Île-de-France, France
Contributed talk: “Creep motion of driven disordered systems”; awarded Best Talk Prize
- June 2026 **Group seminar, Institut Lumière Matière, Modélisation de la Matière Condensée et des Interfaces**, Lyon, Auvergne-Rhône-Alpes, France
Invited seminar: “Two length scales in finite-temperature creep of an elastic interface”

- May 2026 **5e Journées du GDR IDE**, Roscoff, Bretagne, France
Contributed talk: "Two length scales in finite-temperature creep of an elastic interface"
- October 2025 **Journée de la Fédération Friedel-Jacquiot**, *Laboratoire de Physique des Solides*, Orsay, Île-de-France, France
Contributed poster: "Creep motion of ferromagnetic domain walls at finite temperature"
- September 2025 **4èmes Journées du GDR IDE**, Sète, Occitanie, France
Contributed talk: "Creep motion of an elastic line driven in a disordered landscape"
- July 2025 **Fundamental Problems in Statistical Physics XVI**, *Oropa Sanctuary*, Biella, Piemonte, Italy
Contributed poster: "Creep motion of ferromagnetic domain walls at finite temperature"

Teaching Activities

- 2025 **Tutoring for Licence Stage**, *Sorbonne Université*, Paris, France
Co-supervision of undergraduate internship on the topic "Fiber Bundle Model: thermally activated failure and creep dynamics", with Damien Vandembroucq (PMMH, ESPCI)
- 2024 **High School Tutoring – Mathematics**, *Liceo Statale "Giuseppe Novello"*, Codogno, Italy
16 hours
- 2024 **High School Teaching – Mathematics and Physics**, *Liceo Statale "Giuseppe Novello"*, Codogno, Italy
19 hours per week for 4 weeks
- 2023 **University Tutoring – Electromagnetism**, *Università degli Studi di Pavia*, Pavia, Italy
15 hours of teaching
Supervisor: Prof. Lucio Andreani

Honors and Awards

- September 2017 **Winner of the national competitive admission**, *Collegio Ghislieri*, Pavia, Italy

Academic Service and Outreach

- 2025–Present **Student Representative**, *Fédération Friedel-Jacquiot Scientific Board*, Université Paris-Saclay, France
- October 3, 2025 **Science Outreach Facilitator**, *Fête de la Science*, Université Paris-Saclay, France
Organization of a workshop for high-school students in French
- 2018–2021 **Student Representative**, *Department Council*, Università degli Studi di Pavia, Pavia, Italy
Elected representative of students enrolled in Bachelor's and Master's degree programs in Physics
- 2013–2017 **Student Representative**, *Class Council*, Liceo Statale "Giuseppe Novello", Codogno, Italy

Language Skills

- Italian Mother tongue
English C1
French B2

Programming Skills

- Proficient C++, Python, Mathematica
Working Linux shell, Julia, ROOT
knowledge

Additional Training

- February 2026 **Conscience et identité : la querelle indienne du soi**, *Collège de France*, Paris, France
8h formation course, Chaire: Isabelle Ratié
- September 2025 **Writing for Publication**, *Université Paris-Saclay – Academic Writing Center*, Paris, France
6-hour training course
- March 2025 **Ruche Training Course (Cluster Computing)**, *Université Paris-Saclay*, Paris, France
4-hour training course
- March 2025 **Object-Oriented C++**, *Université Paris-Saclay*, Paris, France
4-hour training course
- 2022 **French – B2**, *Università degli Studi di Pavia*, Pavia, Italy
Instructor: Pauline Armellini
- 2021 **French – B1**, *Università degli Studi di Pavia*, Pavia, Italy
Instructor: Pauline Armellini
- 2020 **Python: an Introduction**, *Collegio Ghislieri*, Pavia, Italy
Instructor: Riccardo Invernizzi
- August 2016 **Cours de civilisation française de la Sorbonne**, *Sorbonne Université*, Paris, France
Instructor: Ariela Sillam